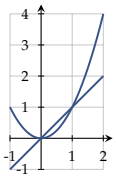
Sheet 24 Tue. 15 Dec. 2026 Solving f $(x) = g(x)$ *et* $f(x) < g(x)$

1. Warm-up · Flash [EQ] Solve 2 $x - 1 = x + 3$.

Ans. $x = 4$.

2. Core · Graph [REG] Graphs of $f(x) = x$ and $g(x) = x^2$:



Solve $f(x) = g(x)$ then $f(x) < g(x)$.

Ans. $x = x^2 \Leftrightarrow x(x-1) = 0 : x = 0$ or $x = 1$; $f(x) < g(x)$: the parabola is above the line for $x < 0$ or $x > 1$.

Intervals: $]-\infty;0[\cup]1;+\infty[$.

3. Core + · Reading code [ALG] What does `f(-1) == g(-1)` ? And `f(0) == g(0)` ?

```
def f(x):
    return x*x
def g(x):
    return 2*x + 3
```

Ans. True $(1 = -2 + 3)$ then False $(0 \neq 3)$: -1 is a solution of $f(x) = g(x)$, not 0.

4. Back · Problem [MOD] Plan A: 15,000 Ar fixed + 1,000 Ar per session. Plan B: 2,500 Ar per session. From how many A sessions is it cheaper?

Ans. $15\,000 + 1\,000x < 2\,500x \Leftrightarrow x > 10$: from 11 sessions on (10: equality at 25,000 Ar).

5. Running thread · Flash [REP] Calculate the coordinates of the midpoint of $[AB]$ with $A(-2;5)$ and $B(6;1)$.

Ans. $\left(\frac{-2+6}{2}; \frac{5+1}{2}\right) = (2;3)$.

One-variable descriptive statistics

Sheet 25 Thu. 17 Dec. 2026 Mean and linearity of the mean

1. Warm-up · Flash [STA] Mean of 12, 15, 9, 14?

Ans. $\frac{50}{4} = 12{,}5$.

2. Core · Method [STA] The data set 4, 6, 8, 10, 12 has mean 8. What is the mean if we add 3 to each value? if each value is multiplied by 2?

Ans. Linearity: $8 + 3 = 11$; $2 \times 8 = 16$.

No need to redo the sum.

3. Core + · True/false [STA] True or false? Justify. The mean of two numbers $a < b$ is always strictly between a and b.

Ans. Right. $a < \frac{a+b}{2} < b$ since $b - a > 0$.

4. Back · Reading [LEC] In a circular diagram of 80 people, one sector represents 25%. What's his angle?

Ans. 20 people; 90 °.

5. Running thread · Flash [DIV] Calculate gcd $(84\,;\,126)$ and deduce the irreducible form of $\frac{84}{126}$.

Ans. GCD $= 42$; $\frac{84}{126} = \frac{2}{3}$.

Sheet 26 Mon. 4 Jan. 2027 Median and quartiles

1. Warm-up · Flash [STA] Order 7, 2, 9, 4, 5, 11, 3 then give the median.

Ans. 2 ; 3 ; 4 ; 5 ; 7 ; 9 ; 11 ; median 5 (4th value).

2. Core · Method [STA] Series 3 ; 5 ; 6 ; 8 ; 9 ; 12 ; 14 ; 15. Give Q_1, median Q_3 and the interquartile range.

Ans. $n = 8$: Q_1 row $\lceil 8/4 \rceil = 2$: 5 ; median $\frac{8+9}{} = 8{,}5$; Q_3 index 6: 12; $Q_3 - Q_1 = 7$.

Convention: the smallest value such that at least 25 % (resp. 75 %) of the data are less than or equal to it.

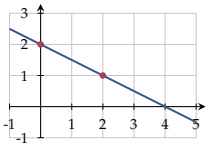
3. Core + · True/false [STA] True or false? Justify. At least half of the values of a series are greater than or equal to its median.

Ans. True. By definition of the median (at least 50 % of the values on each side, endpoints included).

4. Back · Flash [CMP] Order 0.6, $\frac{5}{8}$ and 61 % in increasing order.

Ans. $0{,}6 < 61\,\% = 0{,}61 < \frac{5}{8} = 0{,}625$.

5. Running thread · Graph → formula [REG] The line below represents an affine function f. Determine $f(x)$.



Ans. It passes through $(0;2)$ and $(2;1)$: slope $\frac{1-2}{2-0} = -\frac{1}{2}$, $f(x) = -\frac{1}{2}x + 2$.

Sheet 27 Tue. 5 Jan. 2027 Standard deviation; effect of adding or removing a value

1. Warm-up · Flash [CAL] Calculate $(3-5)^2 + (7-5)^2$.

Ans. $4 + 4 = 8$.

2. Core · Method [STA] Series 2 ; 4 ; 6 : mean, variance and standard deviation (approximate value to 10^{-2}).

Ans. Mean 4; variance $\frac{4+0+4}{3} = \frac{8}{3}$; standard deviation $\sqrt{\frac{8}{3}} \approx 1{,}63$.

3. Core + · True/false [STA] True or false? Justify. Adding 5 to each value of a series does not change its standard deviation.

Ans. True. All the values and the mean are shifted by 5: the deviations from the mean are unchanged.

4. Back · Construction steps [CON] Let ABC a triangle, I the midpoint of $[AB]$ and D the reflection of C with respect to I. What type of $ACBD$?

Ans. A parallelogram: its diagonals $[AB]$ and $[CD]$ have the same midpoint I.

5. Running thread · Flash [GEO] $(MN) \parallel (BC)$, $M \in [AB]$, $N \in [AC]$, $AM = 3$, $AB = 5$, $AC = 10$. Calculate AN.

Ans. Thales: $\frac{AN}{AC} = \frac{AM}{AB}$, therefore $AN = 10 \times \frac{3}{5} = 6$.

Sheet 28 Thu. 7 Jan. 2027 Comparison of series

1. Warm-up · Flash [STA] Range of 3, 8, 5, 12?

Ans. $12 - 3 = 9$.

2. Core · Interpret [STA] Two series A and B have the same mean 12; $\sigma_A = 1.5$ and $\sigma_B = 4$. Which is the more regular?

Ans. A : the values are more tightly clustered around the mean (σ smaller).

3. Core + · Interpret [STA] A : median 11, interquartile range 2. B : median 13, interquartile range 6. Compare the two series.

Ans. B is centred higher (median 13 > 11) but more spread out (interquartile range 6 > 2).

4. Back · Flash [PRB] Two fair coins are tossed. Probability of getting two tails?